

A) [10 Marks] State whether each of the following statements is True or False.

1. When filtering an image with a Gaussian filter, the filter size is the most important because it defines the smoothing scale.
2. Averaging is considered an intuitive solution for image smoothing as it assumes the independence of the noise added to the image.
3. Linear filtering complexity is dependent on both image size and filter size.
4. Sobel filters are typically used as edge detectors rather than sharpening tools.
5. Canny edge detector is based on the 2nd derivative of a Gaussian.
6. Given the following filter, both convolution and correlation operations will be similar.

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

7. The Median filter will perform better in removing noise from the following image.



8. It is important to convolve an image with a Gaussian filter before convolving with a Laplacian filter.
9. For finding horizontal edges in an image, we use the following filter
10. In canny edge detector, reducing both the high and low thresholds of the threshold hysteresis step, will get more discontinuous edges.

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$

B) [3 Marks] In the Harris feature detector, given the following second-moment matrix $M =$

$$\begin{bmatrix} 403 & 385 \\ 385 & 381 \end{bmatrix}, \text{ compute the Harris response function } R \text{ with } \alpha = 0.04.$$

Do you think that this value indicates the existence of a corner, edge or a flat area?

C) [2 Marks] Consider the following 4 matrices are the output of applying the Difference of Gaussians (DoG) on images of the scale space in the SIFT algorithm. There is a single SIFT keypoint that exists in this scale. Your job is to find it. Apply the Keypoint localization step on these matrices to detect the keypoint.

- Report its location in terms of x , y and the scale number it exists in.
- Explain why is this a SIFT keypoint.

0	2	0	1
0	-2	0	1
1	2	0	1
2	2	1	1

Scale =1

1	2	0	2
0	-2	0	1
1	1	0	2
1	1	1	1

Scale=2

2	3	0	2
1	-3	1	1
2	0	1	1
1	1	2	1

Scale=3

2	5	0	4
4	-2	8	6
0	8	-2	3
-1	1	2	2

Scale =4

Difference of Gaussian (DoG) images